

## DAY 3 SWITCH & ROUTER CONFIGURATION BASICS THEORY

Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

ANS:

**Step 1:** Open Cisco Packet Tracer, place a Switch (e.g., 2960), and click on it to open the CLI (Command Line Interface) tab.

**Step 2:** Enter global configuration mode and create the VLANs by typing:

Plaintext

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# vlan 20
```

```
Switch(config-vlan)# name Support
```

```
Switch(config-vlan)# end
```

**Step 3:** Verify your configuration by typing `show vlan brief` to confirm both 'Sales' (ID 10) and 'Support' (ID 20) are active.

Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.<br><br><em><strong>Hint:</strong> Use the interface range command to select multiple ports efficiently.</em>

ANS:

**Step 1:** Enter interface configuration mode for your ports on the switch:

Plaintext

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# interface range fa0/1 - 3
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

**Step 2:** Configure the ports for VLAN 20:

Plaintext

```
Switch(config)# interface range fa0/4 - 5
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# end
```

**Step 3:** Verify the port assignments by typing `show vlan brief` to confirm ports `Fa0/1` through `Fa0/3` are under **Sales (10)** and `Fa0/4` through `Fa0/5` are under **Support (20)**.

Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

ANS:

**Step 1:** Configure the interfaces connecting the two routers with IP addresses:

Plaintext

```
Router1> enable
```

```
Router1# configure terminal
```

```
Router1(config)# interface gigabitethernet 0/0
```

```
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router1(config-if)# no shutdown
```

```
Router1(config)# interface serial 0/1/0
```

```
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
```

```
Router1(config-if)# no shutdown
```

**Step 2:** Configure a static route on Router 1 to reach Router 2's LAN network (`192.168.2.0/24`) via the serial link next-hop IP (`10.0.0.2`):

Plaintext

```
Router1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

```
Router1(config)# end
```

**Step 3:** Verify your routing table by running `show ip route` on Router 1 to confirm the static (S) route entry is present.

Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. <br><br><em><strong>Hint: </strong> Think about router-on-a-stick or Layer 3 switch concepts.</em>

ANS:

**Why the Ping Fails:**

- VLANs (Sales VLAN 10 and Support VLAN 20) are isolated broadcast domains operating at Layer 2.

- Switches only forward traffic within the same VLAN; they cannot route packets between different subnets without a Layer 3 device.

#### How to Enable Inter-VLAN Communication:

- **Router-on-a-Stick:** Connect the switch to a router using a trunk port, and configure subinterfaces on the router (e.g., `G0/0.10` for VLAN 10 and `G0/0.20` for VLAN 20) using 802.1Q encapsulation to act as default gateways.
- **Layer 3 Switch:** Enable routing on a multilayer switch (`ip routing`) and configure Switch Virtual Interfaces (SVIs) for each VLAN (`interface vlan 10` and `interface vlan 20`) with respective IP addresses.